REASEARCH PROJECT RISK ASSESSMENT FORM

SCHOOL OF SCIENCE AND ENGINEERING

August 2025, V5

This form must be completed prior to the commencement of work to ensure that a suitable and sufficient assessment of reasonably foreseeable risks to health and safety is made, and appropriate reasonably practicable controls identified and implemented.

Individuals working under this risk assessment have a legal responsibility to ensure they follow the control measures stipulated to safeguard the health and safety of themselves and others.

The guidance document for completing this form should be consulted here

SECTION 1

OUTLINE THE PROJECT (complete well in advance of the project)

Title of the project: Boltzmann sampling
Department: Mathematics
Location(s) of work (including fieldwork): School of Mathematics and Statistics, Central South University
Description of the project, including methods, fieldwork locations & duration: This project focuses on the theoretical development and practical application of Boltzmann sampling. Boltzmann sampling is a powerful algorithmic paradigm used for generating uniform random samples of structured objects (such as trees, graphs, sequences, tilings, etc.). Method: The project will be carried out in three key stages: 1. Theoretical foundation research: In-depth exploration of the basic principles of Boltzmann sampling, including analyzing the generating functions of specified combinatorial classes. The focus is on understanding the relationship between the generating functions of the structure and the resulting probability distribution. 2. Algorithm design and implementation: We will design and implement recursive and iterative Boltzmann samplers for selected case studies (such as binary trees, partitions, specific molecular structures). We will use advanced programming languages (such as Python or C++) for implementation to ensure the efficiency and clarity of the algorithm. 3. Testing and verification: Statistical tests will be conducted on the generated samples to verify their uniformity or whether they conform to the expected Boltzmann distribution. This will include comparing the empirical distribution with the theoretical expectation and analyzing the scalability of the algorithm. Fieldwork location:

As a theoretical and computational mathematics project, the main "fieldwork" is conducted in a computational environment. All development, simulation, and analysis will be carried out using the computing resources and software laboratory of the School of Mathematics and Statistics at Central South University. No external physical fieldwork is required. Duration: The project is planned to last for six months to ensure sufficient time for in-depth research, algorithm development, rigorous testing, and final result organization.
Serial No.: Jea-33

SUPERVISOR RESPONSIBLE FOR MANAGING THIS PROJECT

Name: Jean Peyen	Position: Lecturer
Department: Mathematics	Date: 16/10/2025

PERSON UNDERTAKING THIS PROJECT

Name: Yuyang Yao	Signature: 姚雨洋 Yuyang Yao.
Date assessment undertaken: 16/10/2025	Position: Student

SECTION 2

PERSONS AT RISK

Identify all categories of individuals who may be affected either directly or indirectly through the work activity and status.

Who is at risk?		Status
Staff	☐	Trained ☐ Competent ☐ Inexperienced ☐ Disability ☐ Supervision required ☐ Lone working ☐
Students	☒	Trained ☒ Competent ☐ Inexperienced ☐ Disability ☐ Supervision required ☐ Lone working ☐
Visitors	☐	Trained ☐ Competent ☐ Inexperienced ☐ Disability ☐ Supervision required ☐ Lone working ☐
Contractors	☐	Trained ☐ Competent ☐ Inexperienced ☐ Disability ☐ Supervision required ☐ Lone working ☐
Public	☐	Trained ☐ Competent ☐ Inexperienced ☐ Disability ☐ Supervision required ☐ Lone working ☐

PROTECTIVE EQUIPMENT REQUIRED

PPE is a last resort and engineering controls (fume cupboard or Local Exhaust Ventilation) should always be used in preference. Consult with your supervisor regarding Personal Protective Equipment required. Insert any EN standard or details such as FFP2, Versaflo RPE, gloves with break through time etc., steel toe capped boots to be worn at all times below. You can use the Chinese equivalent standards if preferred for example: Chinese national product standard ***GB 24541-2022 – Hand protection – Protective gloves against mechanical risks****. This requires wearing Gloves with break through time; another standard* ***GB 21148-2020-Foot Protection-Safety footwea****r, which requires wearing steel-toe-capped boots; for the mask, FFP2 in the EU standard is equivalent to the Chinese standard KN95, as specified in Standard* ***GB 2626-2019*** *Respiratory Protection - Non-Powered Air-Purifying Particle Respirator.*

Engineering control/s	**N/A**
PPE required (head, eye, ear, mask type, glove etc. Training requirements?	**N/A**

ENVIRONMENTAL RISKS

Possible risks to the environment either from the project or as an accidental release to the environment

N/A

WASTE

Is waste generated in this project:	Type of waste and correct disposal:Click here to enter text.
Yes ☐ No ☒	Chemical Waste Form

SECTION 3

IDENTIFICATION OF PROJECT HAZARDS & EVALUATION OF RISKS

Likelihood harm will occur

1. **Rare** - hasn't happened but could / once every 10 years
2. **Unlikely** - could happen / known to happen / once a year
3. **Possible** – could easily happen / once a month
4. **Likely** – could happen / once a week
5. **Almost certain** – occurs several times a day / daily

Severity of harm

1. **No injury, damage**
2. **Minor injury, first aid required**
3. **Injury requires doctor or hospital attendance / absence**
4. **Major injury resulting in permanent damage**
5. **Fatality**

Risk Level Matrix

	S1	S2	S3	S4	S5
L1	1	2	3	4	5
L2	2	4	6	8	10
L3	3	6	9	12	15
L4	4	8	12	16	20
L5	5	10	15	20	25

Risk Level	Risk Evaluation
1 to 5	**LOW** - No Further action necessary, but if something can easily be done to reduce further then do. Monitor all existing controls.
6 to 9	**MEDIUM** – Further controls must be considered, and situation must be monitored.
10 to 25	**HIGH –** STOP Activity then make immediate improvements- Contact Safety Advisor.

From your proposal above consider main hazards below: (including chemicals)	From your persons at risk (1.3) what actions might cause harm? Think about HOW harm might occur. (Not injury)	List any existing controls in place now.	Risk level (L x S)	Identify any additional controls required (including supervision and training). Cross reference with action register below	New risk level (L x S)
Long-term computer usage in a non-ergonomic manner	Students may suffer from eye fatigue and neck discomfort sustain and static postures due to prolonged coding/debugging.	Take regular breaks and wear blue light-blocking glasses	L1 x S1 = 1		
Enduring a high-pressure crunch period right before a deadline.	A frantic, last-minute rush to meet a deadline creates immense mental turmoil.	Mandated short breaks combined with regular task rotation will be adopted to reduce physical and mental exhaustion.			

From your proposal above consider main hazards below: (including chemicals)	From your persons at risk (1.3) what actions might cause harm? Think about HOW harm might occur. (Not injury)	List any existing controls in place now.	Risk level (L x S)	Identify any additional controls required (including supervision and training). Cross reference with action register below	New risk level (L x S)

Action Register

Action register List all additional controls required	Person responsible for implementing	Supervisor responsible	Timescale	Completed
None required				

SECTION 4

SUMMARY OF SIGNIFICANT FINDINGS

Please summarise; an evaluation of the risks involved; the precautions that are in place (or should be taken) and identify any actions which would improve the control of the identified risks. State whether there is a requirement for a written COSHH form and/or a Standard operating procedure to ensure correct safety measures are followed.

Insert links to relevant COSHH or Risk assessments:	None required
Insert links to relevant SOP	None required
Summary of assessment and required actions: Click here to enter text.	Click here to enter text.

SECTION 5

AGREEMENT TO THE TERMS OF THE PROJECT RISK ASSESSMENT

I confirm that, to the best of my knowledge, this project risk assessment describes all the significant risks associated with this project as well as any appropriate controls to reduce the risk to a level as low as is reasonably practicable. I agree to review this project assessment if there is an incident or changes to the activity/ies described above:

Title	Name	Signature	Date
Project Supervisor	Jean Peyen		16/10/2025
Student	Yuyang Yao	姚雨洋 Yuyang Yao.	16/10/2025